

CHAPTER 20

investment lessons from blackjack and gambling



The aim of this Chapter...

... is to gain as much insight as possible from the world of gambling before embarking on the more 'sophisticated' mathematics of portfolio management.

In this Chapter...

- the rules of blackjack
- blackjack strategy and card counting
- the Kelly criterion and money management
- no arbitrage in horse racing

20.1 INTRODUCTION

When I lecture on portfolio management and the mathematics of investment decisions I often start off with a description of the card game blackjack. It is a very simple game, one that most people are familiar with, perhaps by the name of pontoon, 21 or *vingt et un*. The rules are easy to remember, each hand lasts a very short time, the game is easily learned by children and could well give them their first taste of gambling. For without this gambling element there is little point in playing blackjack.

Since the rules are simple and the probabilities can be analyzed, blackjack is also the perfect game to learn about risk and return, and money management and, perhaps most importantly, to help you learn what type of gambler you are. Are you risk averse or a risk seeker? An important question for anyone who later will work in banking and may be gambling with OPM, other people's money.

Despite blackjack being perfect for learning the basics of financial risk and return, and despite bank training managers liking the idea of people being trained in risk management via this game, I am always asked by those training managers to change the title of my lecture. 'You can't call your lecture "Investment Lessons from Blackjack and Gambling," we'll get into trouble with [regulator goes here].' This is a bit silly. Anyone who doesn't think that investment and gambling share the same roots is silly. I can even go so far as to say that most professional gamblers that I know have a better understanding of risk, return and money management than most of the risk managers I know.

In this chapter we will see some of the ideas that these professional gamblers use.

20.2 THE RULES OF BLACKJACK

Players at blackjack sit around a kidney-shaped table, with the dealer standing opposite. A bird's eye view is shown in Figure 20.1.



Figure 20.1 Blackjack table layout.

Before any cards are dealt, the player must place his bet in front of his table position. The dealer deals two cards to each of the players, and two to himself (one of the dealer's cards is dealt face up and the other face down). This is the state of play shown in the figure. Court cards (kings, queens and jacks) count as 10, ace counts as either one or 11 and all other cards are counted at their face value. The value of the ace is chosen by the player.

The aim of the game for the player is to hold a card count greater than that of the dealer without exceeding 21 (going 'bust').

If the player's first two cards are an ace and a 10-count card he has what is known as 'blackjack' or a natural. If he gets a natural with his first two cards the player wins, unless the dealer also has a natural, in which case it is a standoff or tie (a 'push') and no money changes hands. A winning natural pays the player 3 to 2.

Working clockwise around the table from his immediate left the dealer asks each player in turn whether they want to hit or stand. 'Hit' means to draw another card. 'Stand' means no more cards are taken. If the player hits and busts, his wager is lost. The player can keep taking cards until he is satisfied with his count or busts.

The player also has other decisions to make.

The player is also allowed to double the bet on his first two cards and draw one additional card only. This is called 'doubling down.'

If the first two cards a player is dealt are a pair, he may split them into two separate hands, bet the same amount on each and then play them as two distinct hands. This is called 'splitting pairs.' Aces can receive only one additional card. After splitting, ace + 10 counts as 21 and not as blackjack. If the dealer's up card is an ace, the player may take insurance, a bet not exceeding one half of his original bet. If the dealer's down card is a 10-count card, the player wins 2 to 1. Any other card means a win for the dealer.

It is sometimes permitted to 'surrender' your bet. When permitted, a player may give up his first two cards and lose only one half of his original bet.

The dealer has no decisions to make. He must always follow very simple rules when it comes to hitting or standing. He must draw on 16 and stand on 17. In some casinos, the dealer is required to draw on soft 17 (a hand in which an ace counts as 11, not one). Regardless of the total the player has, the dealer must play this way.

In a tie no money is won or lost, but the bet stays on the table for the next round.

Rules differ subtly from casino to casino, as do the number of decks used.

The casino has an advantage over the player and so, generally speaking, the casino will win in the long run. The advantage to the dealer is that the player can go bust, losing his bet immediately, even if the dealer later busts. This asymmetry is the key to the house's edge. The key to the player's edge, which we will be exploiting shortly, is that he can vary both his bets and his strategy. The first published strategy for winning at blackjack was published by Ed Thorp in 1962 in his book *Beat the Dealer*. In this book Professor Thorp explained that the key ingredients to winning at blackjack were

- the strategy: knowing when to hit or stand, doubledown, etc. — this will depend on what cards you are holding and the dealer's upcard
- information: knowing the approximate makeup of the remaining cards in the deck, some cards favor the player and others the dealer
- money management: how to bet, when to bet small and when to bet large.

20.3 BEATING THE DEALER

The first key is in having the optimal strategy. That means knowing whether to hit or stand. You're dealt an 8 and a 4 and the dealer's showing a 6, what do you do? The optimal strategy involves knowing when to split pairs, double down (double your bet in return for only taking one extra card), or draw a new card. Thorp used a computer simulation to calculate the best strategies by playing thousands of blackjack hands. In his best-selling book *Beat the Dealer* Thorp presented tables like the one in Figure 20.2 showing the best strategies.

But the optimal strategy is still not enough without the second key.

You've probably heard of the phrase 'card counter' and conjured up images of Doc Holliday in a 10-gallon hat. The truth is more mundane. Card counting is not about memorizing entire decks of cards but keeping track of the type and percentage of cards remaining in the deck during your time at the blackjack table. Unlike roulette, blackjack has 'memory.' What happens during one hand depends on the previous hands and the cards that have already been dealt out.

A deck that is rich in low cards, 2s to 6s, is good for the house. Recall that the dealer must take a card when he holds 16 or less, the high frequency of low-count cards

		DEALER'S UP CARD										
		2	3	4	5	6	7	8	9	10	A	
YOUR HAND	17+	S	S	S	S	S	S	S	S	S	S	
	16	S	S	S	S	S	H	H	H	H	H	
	15	S	S	S	S	S	H	H	H	H	H	
	14	S	S	S	S	S	H	H	H	H	H	
	13	S	S	S	S	S	H	H	H	H	H	
	12	H	H	S	S	S	H	H	H	H	H	
	11	D	D	D	D	D	D	D	D	D	H	
	10	D	D	D	D	D	D	D	D	H	H	
	9	H	D	D	D	D	H	H	H	H	H	
	5-8	H	H	H	H	H	H	H	H	H	H	
	A, 8-10	S	S	S	S	S	S	S	S	S	S	
	A, 7	S	D	D	D	D	S	S	H	H	H	
	A, 6	H	D	D	D	D	H	H	H	H	H	
	A, 5	H	H	D	D	D	H	H	H	H	H	
	A, 4	H	H	D	D	D	H	H	H	H	H	
	A, 3	H	H	H	D	D	H	H	H	H	H	
	A, 2	H	H	H	D	D	H	H	H	H	H	
	A, A; 8, 8	SP	SP	SP	SP	SP	SP	SP	SP	SP	SP	
	10, 10	S	S	S	S	S	S	S	S	S	S	
	9, 9	SP	SP	SP	SP	SP	S	SP	SP	S	S	
	7, 7	SP	SP	SP	SP	SP	SP	H	H	H	H	
	6, 6	SP	SP	SP	SP	SP	H	H	H	H	H	
	5, 5	D	D	D	D	D	D	D	D	H	H	
	4, 4	H	H	H	SP	SP	H	H	H	H	H	
	3, 3	SP	SP	SP	SP	SP	SP	H	H	H	H	
	2, 2	SP	SP	SP	SP	SP	SP	H	H	H	H	

Figure 20.2 The basic blackjack strategy.

increases his chance of getting close to 21 without busting. For example, take out all the 5s from a single deck and the player has an advantage of 3.3%! On the other hand, a deck rich in 10-count cards (10s and court cards) and aces is good for the player, increasing the chances of either the dealer busting or the player getting a natural (21 with two cards) for which he gets paid at odds of 3 to 2. In the simplest case, card counting means keeping a rough mental count of the percentage of aces and 10s, although more complex systems are possible for the really committed. When the deck favors the player he should increase his bet, when the deck is against him he should lower his bet. (And this bet variation must be done sufficiently subtly so as not to alert the dealers or pit bosses.)

One of the simplest card-counting techniques is to perform the following simple calculation in your head as the cards are being dealt. With a fresh deck(s) start from zero, and then for every ace and 10 that is dealt subtract one, for every 2–6 add one. The larger the count, divided by an estimate of the number of cards left in the deck, the better are your chances of winning. You perform this mental arithmetic as the cards are being dealt around the table.

In *Beat the Dealer*, Ed Thorp published his ideas and the results of his ‘experiments.’ He combined the card-counting idea, money management techniques (such as the Kelly criterion, below) and the optimal play strategy to devise a system that can be used by anyone to win at this casino game. ‘The book that made Las Vegas change the rules,’ as it says on the cover, and probably the most important gambling book ever, was deservedly in the *New York Times* and *Time* bestseller lists, selling more than 700,000 copies.

Passionate about probability and gambling, playing blackjack to relax, however, even Ed himself could not face the requirements of being a professional gambler. ‘The activities weren’t intellectually challenging along that life path. I elected not to do that.’

Once on a film set, Paul Newman asked him how much he could make at blackjack. Ed told him \$300,000 a year. ‘Why aren’t you out there doing it?’ Ed’s response was that he could make a lot more doing something else, with the same effort, and with much nicer working conditions and a much higher class of people. Truer words were never spoken. Ed Thorp took his knowledge of probability, his scientific rigor and his money management skills to the biggest casino of them all, the stock market.

20.3.1 Summary of winning at blackjack

- If you play blackjack with no strategy you will lose your money quickly. If your strategy is to copy the dealer’s rules then there is a house edge of between 5 and 6%.
- The best strategy involves knowing when to hit or stand, when to split, double down, take insurance, etc. This decision will be based on the two cards you hold and the dealer’s face up card. If you play the best strategy you can cut the odds down to about evens.
- To win at blackjack takes patience and the ability to count cards.
- If you follow the optimal strategy and simultaneously bet high when the deck is favorable, and low otherwise, then you will win in the long run.

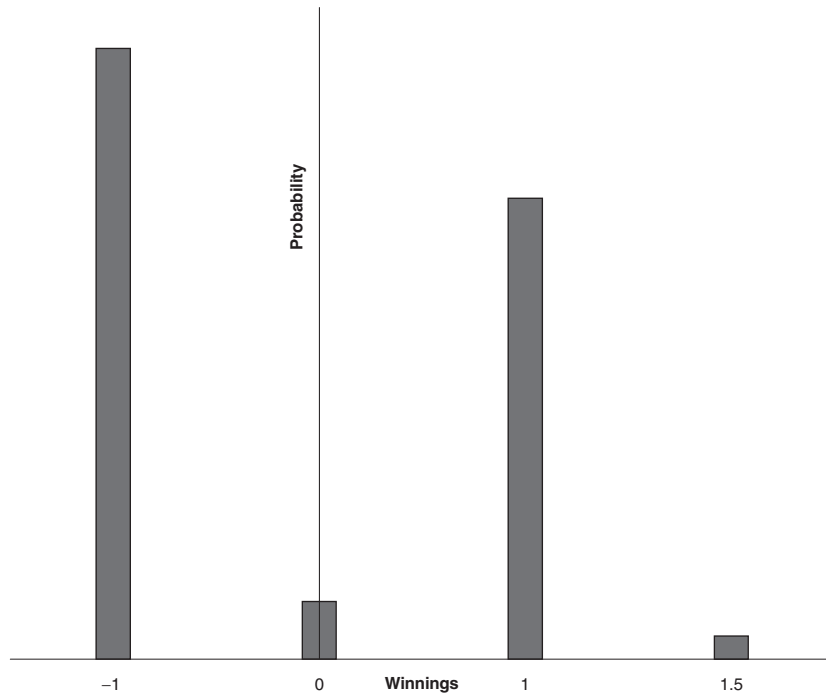


Figure 20.3 The blackjack probability density function (schematic).

What does this have to do with investing?

Over the next two sections we will see how to use estimates of the odds (from card counting in blackjack, say, or statistical analysis of stock price returns) to optimally manage our money.

20.4 THE DISTRIBUTION OF PROFIT IN BLACKJACK

Introduce some notation for the distribution of winnings at blackjack. ϕ is a random variable denoting the outcome of a bet. There will be probabilities associated with each ϕ . Suppose that μ is the mean and σ the standard deviation of ϕ .

In blackjack ϕ will take discrete values:

$$\phi = -1, \quad \text{player loses,}$$

$$\phi = 0, \quad \text{a 'push,'}$$

$$\phi = 1, \quad \text{player wins,}$$

$$\phi = 3/2, \quad \text{player gets a 'natural.'}$$

The distribution is shown (schematically) in Figure 20.3.

20.5 THE KELLY CRITERION

To get us into the spirit of asset choice and money management, consider the following real-life example. You have \$1000 to invest and the only investment available to you is in a casino playing blackjack.

If you play blackjack with no strategy you will lose your money quickly. The odds, as ever, are in favor of the house. If your strategy is to copy the dealer's rules then there is a house edge of between 5 and 6%. This is because when you bust you lose, even if the dealer busts later. There is, however, an optimal strategy. The best strategy involves knowing when to hit or stand, when to split, double down, take insurance (pretty much never), etc. This decision will be based on the two cards you hold and the dealer's face up card. If you play the best strategy you can cut the odds down to about evens, the exact figure depending on the rules of the particular casino.

To consistently win at blackjack takes two things: patience and the ability to count cards. The latter only means keeping track of, for example, the number of aces and 10-count cards left in the deck. Aces and 10 left in the deck improve your odds of winning. If you follow the optimal strategy and simultaneously bet high when there are a lot of aces and 10 left, and low otherwise, then you will do well in the long run. If there are any casino managers reading this, I'd like to reassure them that I have never mastered the technique of card counting, so it's not worth them banning me. On the other hand, I always seem to win, but that may just be selective memory.

The following is a description of the **Kelly criterion**. It is a very simple way to optimize your bets or investments so as to maximize your long-term average growth rate. This is the subject of *money management*. This technique is not specific to blackjack, although I will continue to use this as a concrete example, but it can be used with any gambling game or investment where you have a positive edge and have some idea of the real probabilities of outcomes. The idea has a long and fascinating history, all told in the book by Poundstone (2005). In that book you will also be able to read how the idea has divided the economics community from the gambling community.

We are going to use the ϕ notation for the outcome of a hand of blackjack, but since each hand is different we will add a subscript. So ϕ_i means the outcome of the i th hand.

Suppose I bet a fraction f of my \$1000 on the first hand of blackjack, how much will I have after the hand? The amount will be

$$1000 (1 + f\phi_1),$$

where the subscript '1' denotes the first hand.

On to the second and subsequent hands. I will consistently bet a constant fraction f of my holdings each hand, so that after two hands I have an amount¹

$$1000 (1 + f\phi_1)(1 + f\phi_2).$$



Easy to simulate

¹ This is not quite what one does when counting cards, since one will change the amount f .

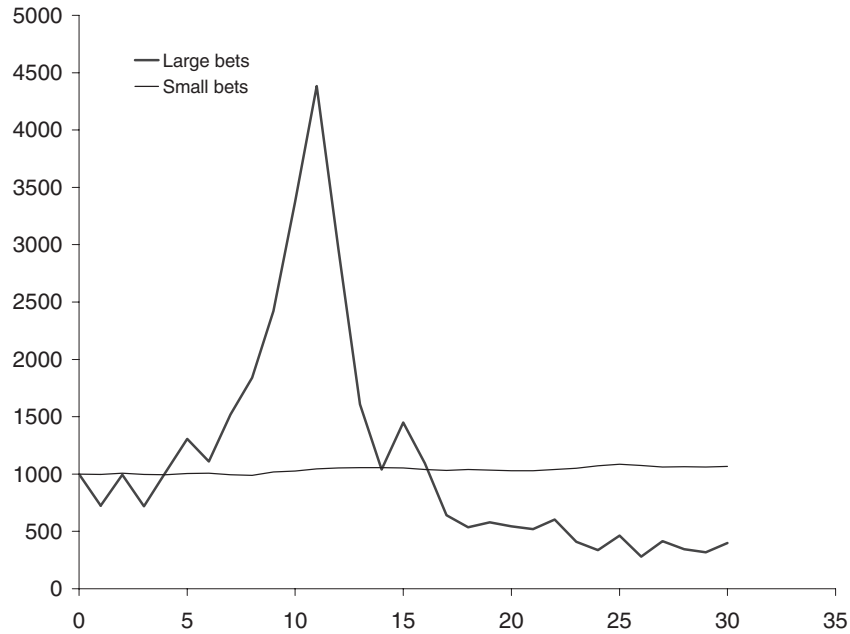


Figure 20.4 Big bets and small bets.

After M hands I have

$$1000 \prod_{i=1}^M (1 + f\phi_i).$$

How should I choose the amount f ?

If f is large, one say, then I will certainly eventually lose all of my money, even if the expected return is positive. If f is very small then it will take me a very long time to make any sizeable profit. These two extremes are demonstrated in Figure 20.4.

Perhaps there is some middle ground where I will make a reasonable profit without risking it all. This is indeed the case, let's do the math.

I am going to choose the fraction f to maximize my expected long-term growth rate. There are other strategies that you can adopt, such as expected wealth maximization. Each goal has its pros and cons.

Expected growth maximization is an obvious goal for a hedge fund since average growth is a number that all potential investors will look at.

This growth rate is given by

$$\frac{1}{M} \log \left(1000 \prod_{i=1}^M (1 + f\phi_i) \right) = \frac{1}{M} \sum_{i=1}^M \log (1 + f\phi_i) + \frac{1}{M} \log(1000).$$

Assuming that the outcome of each hand is independent,² then the expected value of this is

$$E [\log (1 + f\phi_i)],$$

ignoring the scaling factor $\log(1000)$.

² An assumption not strictly true for blackjack, of course.

Expanding the argument of the logarithm in Taylor series, we get

$$E\left[f\phi_i - \frac{1}{2}f^2\phi_i^2 + \dots\right].$$

Now assuming that the mean is small but that the standard deviation is not, we find that the expected long-term growth rate is approximately

$$f\mu - \frac{1}{2}f^2\sigma^2. \quad (20.1)$$

This is maximized by the choice

$$f^* = \frac{\mu}{\sigma^2},$$

giving an expected growth rate of

$$\frac{\mu^2}{2\sigma^2}$$

per hand.

If $\mu > 0$ then $f > 0$ and we stand to make a profit, in the long term. If $\mu < 0$, as it is for roulette or if you follow a naive blackjack strategy, then you should invest a negative amount, i.e. own the casino. (If you must play roulette, put all your money you would gamble in your lifetime on a color, and play once. Not only do you stand an almost 50% chance of doubling your money, you will gain an invaluable reputation as a serious player.)

The long-run growth rate maximization and the optimal amount to invest is called the Kelly criterion.

In Figure 20.5 is shown the function given in Equation (20.1) for the expected long-term growth rate. This example uses $\mu = 0.01$ and $\sigma = 1$. In this figure you will see the optimal

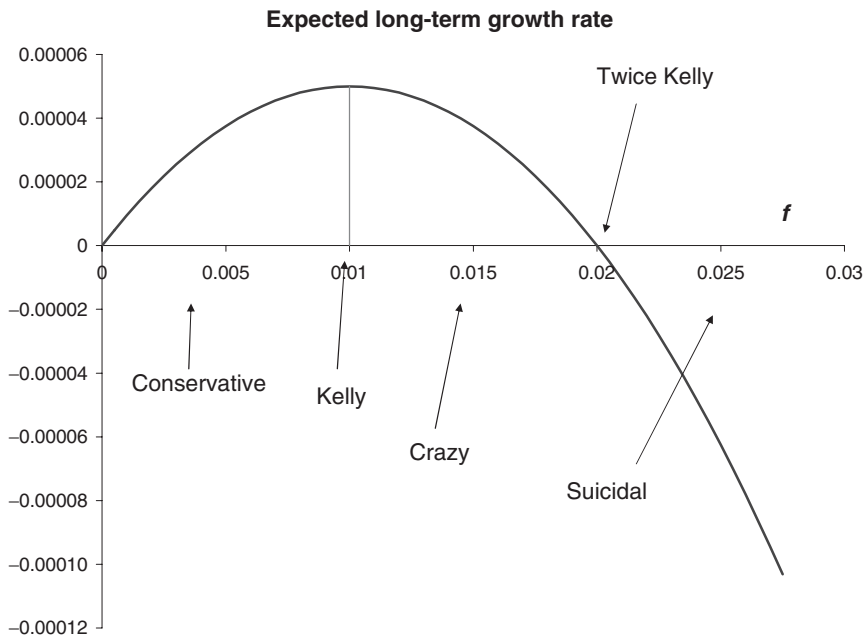


Figure 20.5 The expected long-term growth rate as a function of investment fraction.

betting fraction is the value for f which maximizes the function, this is the Kelly criterion. To the left of this is conservative betting, to the right is crazy. I say crazy because going beyond the optimal fraction increases your volatility of winnings, and decreases their expectation. At 'twice Kelly' your expected winnings are zero, and beyond that they become negative.

Given that in practice you rarely know the odds accurately it makes sense to bet conservatively, in case you accidentally stray into the crazy zone. For that reason many people use 'half Kelly,' that is a fraction that is half of the Kelly fraction. This halves your volatility, keeps you nicely away from the crazy zone, yet only decreases your expected growth rate by 25%.

If you can play M times in an evening you would expect a total growth of

$$\frac{\mu^2 M}{2\sigma^2} \quad (20.2)$$

using full Kelly.



This illustrates one possible way of choosing a portfolio, which asset to invest in (blackjack) and how much to invest (f^*). Faced with other possible investments, then you could argue in favor of choosing the one with the highest (20.2), depending on the mean of the return, its standard deviation and how often the investment opportunity comes your way. These ideas are particularly important to the technical analyst or

chartist who trades on the basis of signal such as golden crosses, saucer bottoms, and head and shoulder patterns. Not only do the risk and return of these signals matter, but so does their frequency of occurrence.

The Kelly criterion is about maximizing expected long-term growth. Many people believe this to be a quite aggressive strategy leading to possible large downturns. There are many other quantities to optimize, of course, so the Kelly criterion might not be the right choice for you. You could, for example, choose to minimize downturn, or maximize some risk-adjusted return, as discussed in the next chapter.

20.6 CAN YOU WIN AT ROULETTE?

Let's go back to Ed Thorp, but a few years before his work on blackjack.

In spring 1955 Ed Thorp was in his second year of graduate physics at UCLA. At tea time one Sunday he got to chatting with colleagues about how to make 'easy money.' The conversation turned to gambling, and roulette in particular. Was it possible to predict, at least with some exploitable degree of accuracy, the outcome of a spin of the wheel? Some of his colleagues, the ones in the know, were certain that the roulette wheels were manufactured so precisely that there were no imperfections that could be discerned, never mind exploited. But Ed's counter to that was simple, if the wheels are so perfect you should be able to predict, using simple Newtonian principles, the path of the ball and its final resting place.

Ed got to work on this problem in the late 1950s, playing around with a cheap miniature roulette wheel, filming and timing the revolutions. He met up with Claude Shannon,

the father of information theory in 1959, originally to discuss his blackjack results, but the conversation soon turned to other games and roulette in particular. Shannon was fascinated. Shortly afterwards they met up at Shannon's house, the basement of which was packed with mechanical and engineering gadgets, the perfect playground for further roulette experiments.

Ed and Shannon together took the roulette analysis to greater heights, investing \$1500 in a full-size professional wheel. They calibrated a simple mathematical model to the experiments, to try to predict the moment when the spinning ball would fall into the waiting pockets. From their model they were able to predict any single number with a standard deviation of 10 pockets. This converts to a 44% edge on a bet on a single number. Betting on a specific octant gave them a 43% advantage.

From November 1960 until June 1961 Ed and Shannon designed and built the world's first wearable computer. The 12 transistor, cigarette-pack sized computer was fed data by switches operated by their big toes. One switch initialized the computer and the other was for timing the rotation of the ball and rotor. The computer predictions were heard by the computer wearer as one of eight tones via an earpiece. (Ed and Shannon decided that the best bet was on octants rather than single numbers since the father of information theory knew that faced with n options individuals take a time $a + b \log(n)$ to make a decision.)

This computer was tested out in Las Vegas in the summer of 1961. But for problems with broken wires and earpieces falling out, the trip was a success. Similar systems were later built for the Wheel of Fortune which had an even greater edge, an outstanding 200%. On 30th May 1985 Nevada outlawed the use of any device for predicting outcomes or analyzing probabilities or strategies.

20.7 HORSE RACE BETTING AND NO ARBITRAGE

Several times we have seen how the absence of arbitrage opportunities leads to the idea of risk-neutral pricing. The value of an option can be interpreted as the present value of the expected payoff, with the expectation being with respect to the risk-neutral asset price path. In this context risk neutral just means that the asset price increases with a growth rate that is the same as the risk-free interest rate. In other words, what we really believe that the asset price is going to do in the future (in terms of its growth rate) is irrelevant. We don't even need to know the growth rate of an asset to price its options, only its volatility.

Something related happens in the world of sports betting.

20.7.1 Setting the odds in a sporting game

In a horse race, football or baseball game the odds are set not to reflect the real probabilities of a horse or a team winning but to reflect the betting that has occurred. Depending on how the betting goes, the odds will be set so that the house/bookie cannot lose. For example, in a soccer match between England and Germany the Germans are more likely to win, but the patriotic English will bet more heavily on England (presumably). The odds given by the bookies will reflect this betting and make it look like England is more likely to win.

Of course, in Germany the situation is reversed. The best bet would be on Germany, but placed in England, and one on England placed in Germany.

In practice bookies in one country would lay off their bets on bookies in other countries so all bookies have roughly the same odds. Otherwise there would be straightforward arbitrage opportunities.

In practice it's unlikely for there to be a sure-fire bet (unless the bookie has made a mistake, the race is fixed, or you can find two or more bookies that aren't directly or indirectly laying off their bets on each other).

But you can win, on average, by exploiting the difference between the real probability of a horse winning and the odds you can get. (There are differences between real odds and what you get paid in all casino games, but it's only in blackjack that this can be exploited.)

20.7.2 The mathematics

Suppose that there are N horses in a race, with an amount W_i bet on the i th horse. The odds set by the bookie are $q_i : 1$. This means that if you bet 1 on horse i you will lose the 1 if the horse loses, but will take home $q_i + 1$ if the horse wins, your original 1 plus a further q_i . How does the bookie set the odds to ensure he never loses?

The total takings before the race are

$$\sum_{i=1}^N W_i.$$

If horse j wins the bookie has to pay out

$$(q_j + 1)W_j.$$

All that the bookie has to do is to ensure that

$$\sum_{i=1}^N W_i \geq (q_j + 1)W_j,$$

or equivalently

$$q_j \leq \frac{\sum_{i=1}^N W_i}{W_j} - 1 \quad \text{for all } j.$$

Nothing too complicated.

But see how the odds have been chosen to reflect the betting. Nowhere was there any mention of the likelihood of horse j winning.

20.8 ARBITRAGE

Suppose the bookie made an error when setting the odds. How could you determine whether there was an arbitrage opportunity? (Don't forget that only positive bets are allowed, there's no going short here.)

Let's introduce some more notation. The w_i are the bets that you place. (We can forget about the wagers made by everyone else, the W s.) Let's assume that your total wager is 1, so that

$$\sum_{i=1}^N w_i = 1. \tag{20.3}$$

The amount you win is

$$(q_j + 1)w_j \quad (20.4)$$

if horse j is the winner.

Can you find a w_i for all i such that they add up to one, are all positive and that expression (20.4) is positive for all j ? If you can there is an arbitrage opportunity.

The requirement that (20.4) is positive can be written as

$$w_j \geq \frac{1}{q_j + 1}. \quad (20.5)$$

Can we find positive w s such that (20.3) and (20.5) hold? This is very easy to visualize, at least when there are two or three horses. Let's look at the two-horse race.

In Figure 20.6 the axes represent the amount of the wager on each of the two horses. The line shows the constraint (20.3). The wagers must lie on this line. The two dots mark the point

$$\left(\frac{1}{q_1 + 1}, \frac{1}{q_2 + 1} \right) \quad (20.6)$$

in each of two situations. One dot is the typical situation where there is no arbitrage opportunity and the other dot does have an associated arbitrage opportunity. Let's see the details.

To find an arbitrage opportunity we must find a pair (w_1, w_2) lying on the line such that each coordinate is greater than a certain quantity, depending on the q s. Plot the point

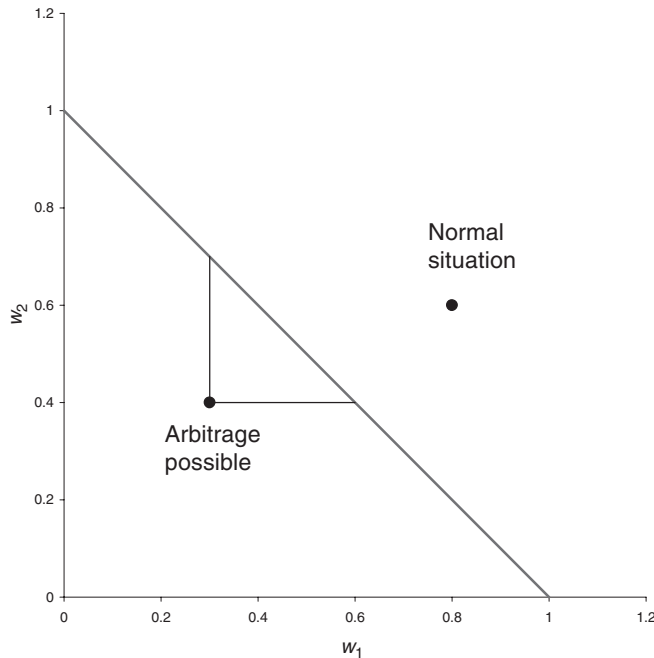


Figure 20.6 Arbitrage in a two-horse race.

(20.6) and draw a line vertically up, and another line horizontally to the right, as shown in the figure, emanating from the dot.

Does the quadrant defined by these two lines include any of the line? If not, as would be the case with the higher dot, then there is no arbitrage possible. If some of the line is included then arbitrage is possible.

20.8.1 How best to profit from the opportunity?

There's a simple test to see whether we are in an arbitrage situation. In general, if

$$\sum_{i=1}^N \frac{1}{q_i + 1} \geq 1$$

then there is no arbitrage. If the sum is less than one, there is an arbitrage.

You can benefit from the arbitrage by placing wagers w_i such that they lie on the part of the line encompassed by the quadrant. Which part of the line, though, is up to you. By that I mean that you must make some statement about what you are trying to achieve or optimize in the arbitrage. One possibility is to look at the worst-case scenario and maximize the payback in that case. Alternatively, specify real probabilities for each of the horses winning.

20.9 HOW TO BET

We saw how odds are established by bookies. We even saw how to spot arbitrage opportunities. In practice, of course, you could spend a lifetime looking for arbitrage opportunities that rarely occur in real life. Now we are going to see if we can exploit the difference between the odds as set by the bookie and the odds that you estimate. Remember, the odds set by the bookie are really determined by the wagers placed, which are more to do with irrational sentiment ('I'm going to bet on this horse 'cos it's got the same name as the pet rat I had when I was a child') than with a cold-hearted estimation of the probabilities.

We need some more notation. Let's use p_i as the probability of the i th horse winning the race. This is supposed to be the real probability, not the bookie's probability. Obviously, the odds must sum to one:

$$\sum_{i=1}^N p_i = 1.$$

If we wager w_i on the i th horse then we expect to make

$$m = \left(\sum_{i=1}^N p_i w_i (q_i + 1) \right) - 1 \quad (20.7)$$

This is under the assumption that the total wager, the sum of all the w s, is one. An obvious goal is to make this quantity positive, we want to get a positive return on average. But there may be many ways to make this positive. How do we decide which way is best?

Table 20.1 Odds and probabilities in a horse race.

Horse	Bookie's odds	Your estimate of probability	Wager
Nijinsky	5	0.2	
Red Rum	6	0.2	
Oxo	1	0.1	
Red Marauder	1	0.1	
Gay Lad	2	0.1	
Roquefort	2	0.1	
Red Alligator	2	0.1	
Shergar	2	0.1	

Another quantity we might want to look at is the standard deviation of winnings. This is given by

$$\sqrt{\sum_{i=1}^N p_i (w_i(q_i + 1) - 1 - m)^2} \quad (20.8)$$

This measures the dispersion of winnings about the average, and is often interpreted as a measure of risk. If this were zero our profit or loss would be a sure thing.

Here's an example, see Table 20.1.

How should you bet? The following calculations are easily done on a spreadsheet.

Scenario 1: Maximize expected return Since you place no premium on reducing risk you should bet everything on the horse that maximizes

$$p_i(q_i + 1).$$

In this case, that is Red Rum. The expected return is 40% with a standard deviation of 280%. A very risky bet. See Table 20.2.

Scenario 2: Minimize standard deviation An interesting strategy.

I say 'interesting' because this strategy results in zero standard deviation, and a return of -62%. In other words, a guaranteed loss! See Table 20.3.

Table 20.2 Maximizing expectation.

Horse	Bookie's odds	Your estimate of probability	Wager
Nijinsky	5	0.2	0
Red Rum	6	0.2	1
Oxo	1	0.1	0
Red Marauder	1	0.1	0
Gay Lad	2	0.1	0
Roquefort	2	0.1	0
Red Alligator	2	0.1	0
Shergar	2	0.1	0

Table 20.3 Minimizing standard deviation.

Horse	Bookie's odds	Your estimate of probability	Wager
Nijinsky	5	0.2	0.063062
Red Rum	6	0.2	0.054068
Oxo	1	0.1	0.189203
Red Marauder	1	0.1	0.189246
Gay Lad	2	0.1	0.126108
Roquefort	2	0.1	0.126108
Red Alligator	2	0.1	0.126108
Shergar	2	0.1	0.126108

Table 20.4 Maximize return divided by standard deviation.

Horse	Bookie's odds	Your estimate of probability	Wager
Nijinsky	5	0.2	0.459016
Red Rum	6	0.2	0.540984
Oxo	1	0.1	0
Red Marauder	1	0.1	0
Gay Lad	2	0.1	0
Roquefort	2	0.1	0
Red Alligator	2	0.1	0
Shergar	2	0.1	0

Scenario 3: Maximize return divided by standard deviation A strategy that seeks to benefit from a positive expectation but with a smaller risk. For mathematical reasons (the Central Limit Theorem) this is a natural strategy. The solution is given in Table 20.4.

The expected return is now 31% with a standard deviation of 164%.

20.10 SUMMARY

The mathematics of gambling is almost identical to the mathematics of 'investing.' The main difference between gambling and investing is that the parameters are usually easier to measure with gambling games. If you can't cope with the mathematics (and the emotional roller coaster ride) of gambling then you shouldn't be working in a bank;-

FURTHER READING

- See Kelly's original 1956 paper.
- The classic reference texts on blackjack are by Thorp (1962) and Wong (1981).
- The gripping story of John Kelly, Claude Shannon, Ed Thorp and a cast of many intriguing characters can be found in Poundstone (2005).

EXERCISES

1. A die is weighted so that the probability of getting a 6 is 0.3, of getting a 1 is 0.1, and of getting 2, 3, 4 or 5 is 0.15 each. You bet on 6, receiving odds of 5 to 1 (fair odds for an unbiased die). What is your expected profit? What is your standard deviation of profit? What is the formula for the expected return as a function of the fraction of your wealth that you bet? What fraction maximizes this expected return?
2. Build a spreadsheet that simulates the above, in which you keep betting the same fraction of your accumulated wealth.

